

Group Project Overview – Flint Water Line Replacements

What is the project about?

The Mayor of the City of Flint has reached out to your analytics team for help in reducing costs for fixing Flint's water crisis. You are tasked with helping to predict areas where water service lines need to be replaced due to dangerous materials. The previous firm was fired in March 2019. The hope is that predictive analysis can help to better target areas of high risk in order to prioritize the infrastructure investments.

What are you supposed to do?

The aim is to predict whether the water service line of a home/parcel of land is likely to need to be replaced. A service line needs to be replaced if either the public or private side has dangerous material (Lead or Galvanized). The target column is "dangerous" (the last column), which has been created using the two included variables `sl_private_type` and `sl_public_type` (for the materials on the private and public side of the service line, respectively). This is a classification problem - each observation needs to be classified as TRUE (1) or FALSE (0).

But what are we to do specifically?

Ultimately, that is up to you. You have been provided with Flint parcel data. Descriptions of the variables are provided in the Appendix. You have been provided a sample of 20,697 records - the pre-March 2019 data. An additional 6,166 parcels await your analysis. They are being held out in a test set. The test set also has a "dangerous" target column; but it has just been randomized. That is, the predictors data will be real, but the target will be junk. Ultimately, your analysis will be evaluated using the true test target that has been held out. The criterion for performance evaluation is the AUC of the ROC.

What are your deliverables?

You will need to provide four specific deliverables:

1. A small pitch presentation document (a Powerpoint document) that you can deliver in 15 mins. Top management of the state can, at least initially, only give you 15 mins of their time and you need to be able communicate your results in that time.
2. An actual presentation made in class, using your slide deck from part 1. All members from the group should aid in delivery of the presentation.
3. A project report (a Word document) that shows your work. More details about the report (page limit, audience, etc.) will be forthcoming.
4. Also, your solution technique will be scored, using a test dataset that has been held out. You must provide detailed instructions on running your analysis. Finally, you need to

provide the files needed to score the test data, and the AUC of the test data with the random/junk target column. You may want to use JMP saved scripts.

Appendix – Variable Descriptions

Data fields

pid – Unique parcel id
Property Zip Code - Zip code of parcel
Owner Type - Whether parcel is privately owned or not.
Owner State - The state that that owner of the parcel lives in
Homestead - Homestead is a person's or family's residence, which comprises the land, house, and outbuildings,
Homestead Percent - 0-100. We don't really know what this is.
HomeSEV - State Estimated Value of home
Land Value - Land Value
Land Improvements Value - Improvements are things like wells, buildings, etc.
Residential Building Value - Value of residential buildings, if any
Residential Building Style - Style of residential building
Commercial Building Value - Value of commercial buildings, if any
Building Storeys - Number of storeys
Parcel Acres - Size of parcel in acres
Rental - Whether the property is a rental or not.
Use Type - Residential, Commercial, or Industrial use
Prop Class - Whether a parcel is agricultural, industrial, residential, or commercial property
Old Prop class - Historical prop class
Year Built - Year home is built
USPS Vacancy - Whether the home had an active post office address
Zoning - Zoning
Future Landuse - Planned use for land in the future
DRAFT Zone - Future assigned zoning
Housing Condition 2012 - 2012 survey of housing conditions
Housing Condition 2014 - Ditto 2014
Commercial Condition 2013 - Ditto for commercial properties
Latitude
Longitude
Hydrant Type - What kind of fire hydrant is nearest to the property
Ward - In which ward the parcel resides.
PRECINCT - Voting precinct
CENTRACT - Census Tract
CENBLOCK - Census Block
SL_Type - Service line type in historical records ([found in a basement](#))
SL_Type2 - Service line type in historical records
SL_Lead – Indicator variable constructed from SL_Type and SL_Type2: if either indicates lead
SL_private_inspection – The result of a home side inspection, if any. The inspection reveals the material of the pipe coming into the home (the private side).

Columns AJ-AM (Last_Test, Max_Lead, Med_Lead, Num_Tests) are water test results in ppb for any water samples associated with the parcel. There may be multiple samples per parcel, so the max, median, mean, and last test results are given.

Note: SL_Type and SL_Type2 are the service line material of the public and private parts of the service line, respectively, according to the historical records (found in a basement). The materials in the service lines correspond to the following values:

- 0: 'Unkown/Other',
- 1: 'Copper',
- 2: 'Tubeloy',
- 3: 'Zinc',
- 4: 'Lead',
- 5: 'Copper/Lead',
- 6: 'Copper/Zinc',
- 7: 'Copper/Tubeloy',
- 8: 'Lead/Zinc',
- 9: 'Lead/Tubeloy',
- 10: 'Copper/?',
- 11: 'Galvanized/Other'